1. Project Overview

This project develops an end-to-end analytics solution to predict telecom customer churn, explain key churn drivers, and present insights via interactive dashboards for business users. The solution spans data ingestion, data cleaning and feature engineering, SQL-based customer behaviour analysis, supervised machine learning models, and self-service BI.

2. Objectives and Scope

2.1 Objectives

- Identify customers most likely to churn using supervised classification models.
- Explain why customers churn through feature importance analysis and model interpretability.
- Provide actionable insights through a Power BI dashboard.

2.2 Scope

- Focus on a telecom customer base with subscription, usage, billing, and support interaction data.

3. Dataset Description

The project uses the Telco Customer Churn dataset (Telcocustomerchurn.xlsx), containing 7,043 records and 33 columns.

3.1 Feature Groups

- **Customer identifiers and geography:** CustomerID, Country, State, City, Zip Code.
- **Demographics:** Gender, Senior Citizen, Partner, Dependents.
- **Subscription and services:** Tenure Months, Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies, Contract.
- **Billing and value:** Paperless Billing, Payment Method, Monthly Charges, Total Charges, CLTV.
- **Target and churn labels:** Churn Label, Churn Value (binary target), Churn Score, Churn Reason.

After cleaning, there are no missing values in any column, including Churn Reason.

4. Solution Architecture

The solution follows the phases defined in the project brief:

1. **Data Source** – Telecom churn data from CSV/Excel and optionally a SQL database.
2. **Data Processing (Python)** – Cleaning, encoding, feature engineering, and EDA in Pandas/NumPy with Matplotlib/Seaborn.
3. **SQL Layer** – Customer behaviour and financial metrics computed in MySQL/PostgreSQL.
4. **Machine Learning Models** – Multiple classifiers (Logistic Regression, Random Forest, XGBoost, CatBoost, Decision Tree) built in scikit-learn, XGBoost, and CatBoost.
5. **Visualisation (Power BI)** – Executive and analytical dashboards for churn, revenue, and behaviour analysis.
6. **GenAI Streamlit App** – Python + LLM-based app to answer natural-language questions over processed data or the SQL layer.

5. Methodology

5.1 Data Cleaning and Feature Engineering (Phase 1)

Data cleaning and feature engineering are implemented in project1_phase1.ipynb.

5.1.1 Loading and Inspection

- Read the Excel file into a Pandas DataFrame.
- Use head(), info(), and missing-value checks to understand structure and quality.

5.1.2 Handling Missing Values and Duplicates

- The only column with missing data is Churn Reason; missing entries are imputed with the value “Retained”, eliminating all nulls.
- Shape before and after duplicate removal remains 7,043 × 33, indicating no duplicates.

5.1.3 Categorical Encoding and Binary Mapping

- Binary columns (for example, Gender, Partner, Dependents, Paperless Billing, and other Yes/No fields) are mapped to 0/1 using a shared mapping (Yes = 1, No = 0, Male = 1, Female = 0) where applicable.

- Non-numeric textual fields that may cause leakage (Churn Label, Churn Score, CLTV, Churn Reason, Lat Long, Country, Count) are preserved in the raw data but dropped at modelling time.

5.1.4 Feature Engineering

- **Tenure groups:** Bucket Tenure Months into four categories: 0–1 year, 1–2 years, 2–4 years, and more than 4 years, to support tenure-based analysis.
- **Average monthly charges:** Compute average monthly charges as Total Charges / Tenure Months (with safeguards for zero tenure) to approximate revenue per month.
- **Contract × Payment interaction:** Create contract payment interaction by concatenating Contract and Payment Method (for example, "Month-to-month + Electronic check").
- **Payment risk category:** Map payment methods into high (Electronic check), medium (Mailed check), and low (Bank transfer automatic, Credit card automatic).

5.1.5 Exploratory Data Analysis (EDA)

- Plot the distribution of Churn Value (0 vs 1) to quantify churn rate.
- Generate a numeric-only correlation heatmap.
- Compute absolute correlations with Churn Value to identify top drivers such as Churn Score, Tenure Months, Dependents, Monthly Charges, Paperless Billing, Senior Citizen, and Partner.

5.1.6 Rule-Based Risk Segmentation

- **High risk:** Short tenure, high monthly charges, month-to-month contracts, electronic check, no Online Security, and no Tech Support.
- **Low risk:** Longer tenure, lower charges, one- or two-year contracts, automatic payments, and presence of Online Security and Tech Support.
- **Medium risk:** Customers not classified as high or low risk.

A pie chart of the risk segment variable illustrates the proportion of customers in each risk category.

5.2 SQL Analysis (Phase 2 – Design)

The processed dataset is to be loaded into a relational database (MySQL or PostgreSQL) to support SQL analytics.

Planned SQL analyses include:

- Monthly churn rate and running churn (using window functions).
- Churn by plan type and by demographic/behavioural segments.
- High-value customer churn and Average Revenue Per User (ARPU).
- Basic customer lifetime value calculations using revenue and tenure.
- Ranking customers by churn risk based on model scores or rule-based segments.

These SQL outputs will feed the Power BI dashboard and the GenAI app as summarised metrics and analytical views.

5.3 Machine Learning Models (Phase 3)

Machine learning is implemented in project1_phase3.ipynb.

5.3.1 Preprocessing for Modelling

- **Target:** Churn Value is used as the binary target variable.
- **Features:** Drop CustomerID, Churn Label, Churn Value, Churn Reason, Churn Score, CLTV, Lat Long, Latitude, Longitude, Country, and Count to avoid leakage and identifier bias.
- Apply one-hot encoding via pd.get_dummies(..., drop_first=True) to categorical features.
- Coerce all feature columns to numeric with pd.to_numeric(errors="coerce").
- Split data into train (70%) and test (30%) using train_test_split with random_state=42.
- Impute any remaining numeric missing values with the training median.
- Standardize features using StandardScaler fitted on the training set and applied to both train and test sets.

5.3.2 Logistic Regression (Baseline)

A Logistic Regression classifier (max_iter=2000, random_state=42) provides the baseline.

- Accuracy: 0.7478
- Precision: 0.5650
- Recall: 0.4065
- F1 score: 0.4728

- ROC–AUC: 0.7646

Top coefficients (by absolute value) indicate that Internet Service_Fiber optic, Paperless Billing_Yes, Gender_Male, Senior Citizen_Yes, several city dummy variables, and Total Charges bins are strong predictors.

5.3.3 Random Forest

A RandomForestClassifier with 100 estimators (n_estimators=100, random_state=42, n_jobs=-1) improves performance over the baseline.

- Accuracy: 0.8027
- Precision: 0.7165
- Recall: 0.4813
- F1 score: 0.5758
- ROC–AUC: 0.8528

Important features include Tenure Months, Monthly Charges, Zip Code, Internet Service_Fiber optic, Contract_Two year, Dependents_Yes, Payment Method_Electronic check, Online Security_Yes, Contract_One year, and Tech Support_Yes.

5.3.4 XGBoost

An XGBClassifier is trained with 100 trees, learning rate 0.1, random_state=42, use_label_encoder=False, and eval_metric='logloss'.

- Accuracy: 0.8069
- Precision: 0.6772
- Recall: 0.5850
- F1 score: 0.6277
- ROC–AUC: 0.8640

The most important features include Internet Service_Fiber optic, Contract_Two year, Contract_One year, Internet Service_No, Dependents_Yes, Tenure Months, Streaming Movies_Yes, Multiple Lines_Yes, Paperless Billing_Yes, and Payment Method_Electronic check.

5.3.5 CatBoost

A CatBoostClassifier (100 iterations, learning rate 0.1, random_state=42, verbose=0) achieves similar performance.

- Accuracy: 0.8022

- Precision: 0.6840
- Recall: 0.5374
- F1 score: 0.6019
- ROC–AUC: 0.8648

Top features by importance are Tenure Months, Dependents_Yes, Contract_Two year, Contract_One year, Internet Service_Fiber optic, Tech Support_No internet service, Monthly Charges, Payment Method_Electronic check, Zip Code, and Paperless Billing_Yes.

5.3.6 Decision Tree

A DecisionTreeClassifier (random_state=42) is used as an interpretable benchmark.

- Accuracy: 0.7596
- Precision: 0.5781
- Recall: 0.5034
- F1 score: 0.5382
- ROC–AUC: 0.6809

Top features include Tenure Months, Internet Service_Fiber optic, Zip Code, Monthly Charges, Dependents_Yes, Payment Method_Electronic check, Multiple Lines_Yes, Gender_Male, and Paperless Billing_Yes.

5.3.7 Model Selection

Model	Accuracy	Precision	Recall	F1 Score	ROC–AUC	Top Predictive Features
Logistic Regression	0.7478	0.5650	0.4065	0.4728	0.7646	Internet Service_Fiber optic, Paperless Billing_Yes, Gender_Male, Senior Citizen_Yes, City dummies, Total Charges bins
Random Forest	0.8027	0.7165	0.4813	0.5758	0.8528	Tenure Months, Monthly Charges, Zip Code, Internet Service_Fiber optic, Contract_Two year, Dependents_Yes, Payment Method_Electronic check,

Model	Accuracy	Precision	Recall	F1 Score	ROC–AUC	Top Predictive Features
						Online Security_Yes, Contract_One year, Tech Support_Yes
XGBoost	0.8069	0.6772	0.5850	0.6277	0.8640	Internet Service_Fiber optic, Contract_Two year, Contract_One year, Internet Service_No, Dependents_Yes, Tenure Months, Streaming Movies_Yes, Multiple Lines_Yes, Paperless Billing_Yes, Payment Method_Electronic check
CatBoost	0.8022	0.6840	0.5374	0.6019	0.8648	Tenure Months, Dependents_Yes, Contract_Two year, Contract_One year, Internet Service_Fiber optic, Tech Support_No internet service, Monthly Charges, Payment Method_Electronic check, Zip Code, Paperless Billing_Yes
Decision Tree	0.7596	0.5781	0.5034	0.5382	0.6809	Tenure Months, Internet Service_Fiber optic, Zip Code, Monthly Charges, Dependents_Yes, Payment Method_Electronic check, Multiple Lines_Yes, Gender_Male, Paperless Billing_Yes

Given the balance between precision and recall, XGBoost and CatBoost are the top candidates, both achieving ROC–AUC around 0.864 on the test set. XGBoost offers slightly higher F1 score and recall, making it a suitable champion model where identifying churners is crucial. CatBoost is a strong alternative when direct handling of categorical variables is preferred.

6. Key Insights and Business Implications

6.1 Drivers of Churn

- Lower tenure is strongly associated with higher churn; long-tenure customers are more stable.
- Month-to-month contracts show higher churn than one- and two-year contracts.
- Fibre-optic internet customers have a higher churn risk than DSL or “No internet” customers.
- Higher monthly charges, paperless billing, and electronic check payment are linked with higher churn.
- Absence of Online Security and Tech Support is associated with a higher risk.
- Household-related variables (Dependents, Partner) exhibit meaningful correlation with churn.

6.2 Recommended Actions

- Implement tenure-based retention campaigns targeting customers in their first 12 months.
- Incentivise migration from month-to-month to one- and two-year contracts.
- Review pricing and discount structures for high-charge, fiber-optic segments.
- Encourage adoption of automatic payment methods over electronic checks.
- Bundle Online Security and Tech Support for high-risk segments.